



Service Bulletin Number: 6506608

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Minimum Flywheel Dimensions for Resurfacing and Reuse

## Minimum Flywheel Dimensions for Resurfacing and Reuse

### Intent

This bulletin covers flywheel resurfacing guidelines, clarifying the amount of material that can be safely removed during machining and the reuse conditions for a machined flywheel.

### Product Affected

- X15 CM2350 X114B
- X15 CM2350 X116B
- X15 CM2350 X123B
- X15 CM2350 X129B
- X15 CM2450 X124B
- X15 CM2450 X134B
- X15 CM2450 X142B
- X15 CM2670 X145B

### Service Instruction

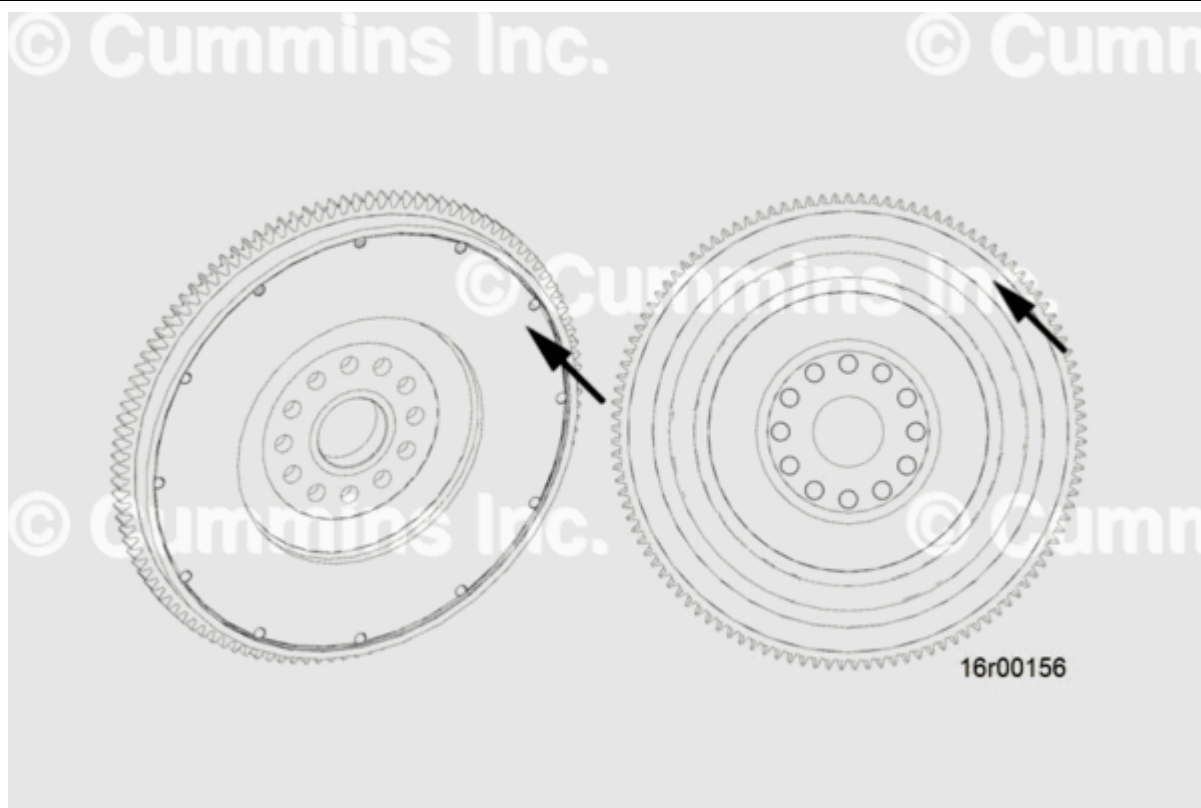


Figure 1, Friction Surface and Mounting Surface Sides of Flywheel. Arrows Show Location to Take Thickness Measurements Using Calipers.

**Table 1, Flywheel Reuse Conditions**

| Flywheel Option | Flywheel Part Number | Flywheel Thickness Requirement After Resurfacing   | Surface Finish Requirement |
|-----------------|----------------------|----------------------------------------------------|----------------------------|
|                 | (ID 16 / ID 10)      |                                                    | (Ra Maximum)               |
|                 |                      |                                                    |                            |
| FW 1020         | 3680922 / 3680921    | 38.75 mm [ 1.52 in ]<br>+/-0.25 mm [ +/- 0.01 in ] | Ra max 3.2                 |
| FW 1038         | 3102508 / 3680921    |                                                    |                            |
| FW 1249         | 3693466 / 3693467    | 48.01 mm [ 1.89 in ]<br>+/-0.25mm [ +/- 0.01 in ]  | Ra max 3.2                 |
| FW 1026         | 3680915 / 3680914    | 44.35 mm [ 1.67 in ]<br>+/-0.25mm [ +/- 0.01 in ]  | Ra max 3.2                 |
| FW 1067         | 3687024 / 3687016    |                                                    |                            |
| FW 1259         | 5575105 / 5575106    |                                                    |                            |

- If the amount of material that will be removed during the machining operation will result in a flywheel thinner than the value listed for the engine's flywheel option in Table 1, do **not** resurface the flywheel.

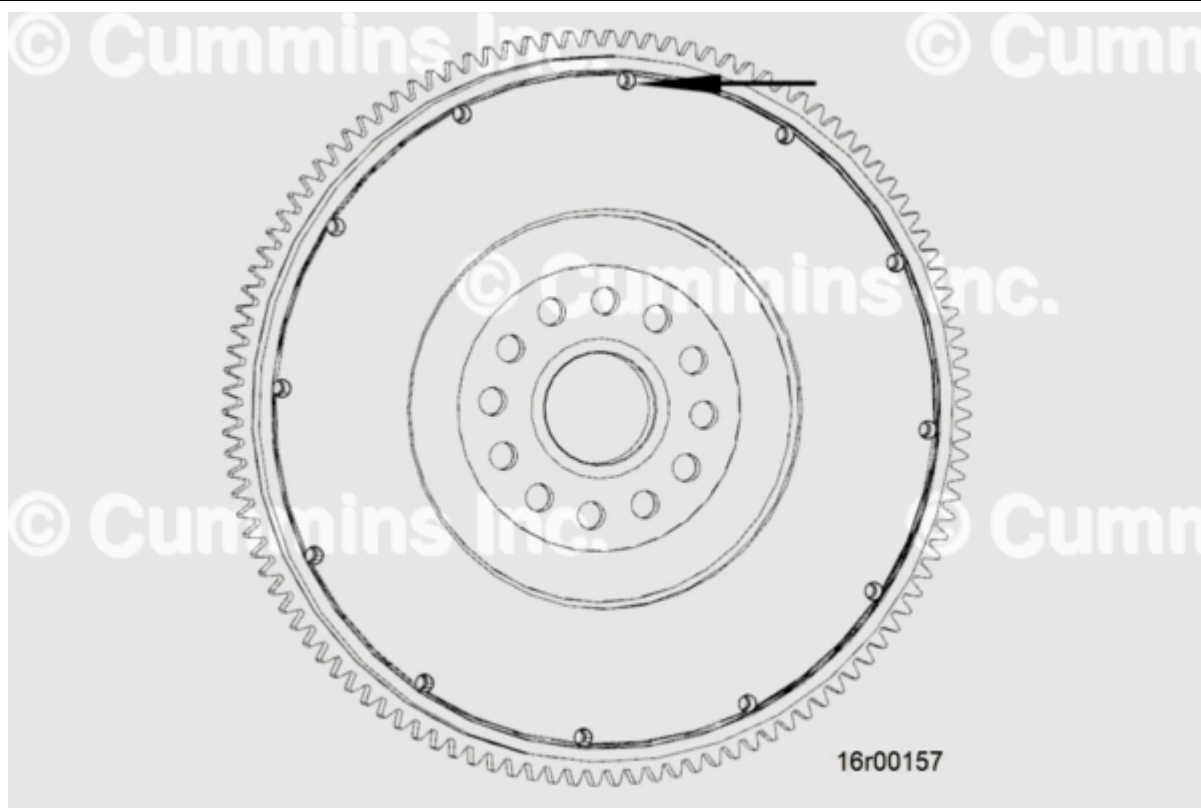


Figure 2, Flywheel Friction Surface and Clutch Mounting Capscrew Holes.

**Table 2, Minimum Clutch Mounting Hole Thread Depth**

| Flywheel Options | Flywheel Part Number | Minimum Clutch Mounting Capscrew Hole Thread Depth | Clutch Mounting Capscrew Hole Thread Pitch |
|------------------|----------------------|----------------------------------------------------|--------------------------------------------|
| FW 1020          | 3680922 / 3680921    | 21.25 mm [ 0.84 in ]                               | 7/16-14 UNC 2B                             |
| FW 1038          | 3102508 / 3680921    |                                                    |                                            |
| FW 1249          | 3693466 / 3693467    | 20.25 mm [ 0.80 in ]                               | M10 x 1.5-6H                               |
| FW 1026          | 3680915 / 3680914    | 20.25 mm [ 0.80 in ]                               | 7/16-14 UNC 2B                             |
| FW 1067          | 3687024 / 3687016    |                                                    |                                            |
| FW1259           | 5575105 / 5575106    | 21.00 mm [ 0.83 in ]                               | M10 x 1.5-6H                               |

- After resurfacing, use a bottoming tap to clean and recut the clutch mounting capscrew hole threads.
- After using a bottoming tap on the threaded holes, measure the thread depth of the clutch mounting capscrew hole with a depth gauge, or micrometer. If the thread depth is less than the minimum thread depth value for the flywheel option the engine uses, listed in Table 2, do **not** reuse the flywheel.

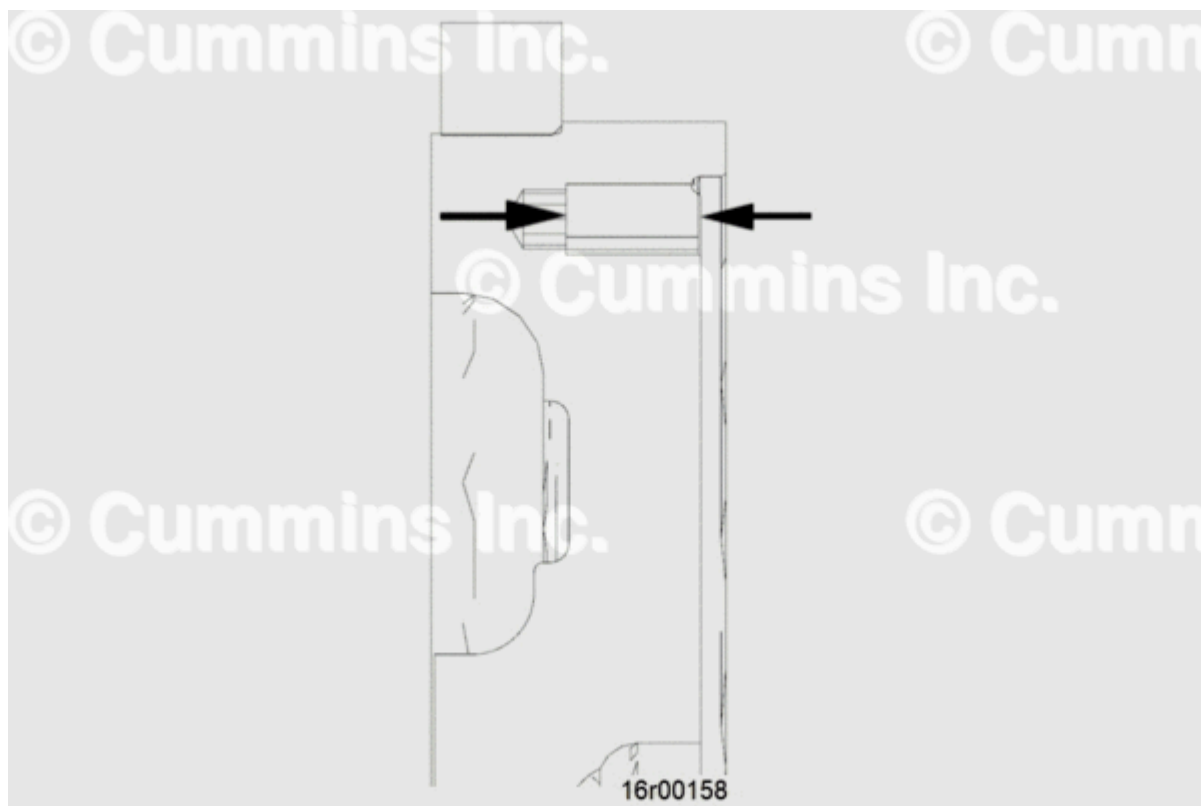


Figure 3, Cutaway Image of Flywheel Showing How Thread Depth Is Measured.

#### Reuse Guidelines

#### **⚠ CAUTION ⚠**

**Neglecting minimum flywheel thickness can result in improper mounting capscrew engagement and clutch damage.**

- After machining the friction surface, measure the thickness of the flywheel to verify that the flywheel is **not** too thin to be reused.
- If more than 1.5 mm [ 0.060 in ] of material is removed from the flywheel friction surface, the clutch disk can contact the flywheel mounting capscrews and cause damage.
- After resurfacing, measure the total runout of the flywheel. If the total runout of the flywheel exceeds the total runout specification for the radius of the engine's flywheel in the engine's service information found on QuickServe Online (QSOL), the flywheel is **not** able to be reused.

#### Publications Affected

| Table 3, Publications Affected |                     |                 |                 |                                                                                   |                   |
|--------------------------------|---------------------|-----------------|-----------------|-----------------------------------------------------------------------------------|-------------------|
| Manual type                    | Engine              | Bulletin Number | Procedure Title | Procedure Number                                                                  | Procedure Section |
| Service                        | X15 CM2350<br>X114B | 5411181         | Flywheel        | Refer to Procedure 016-005 (/qs3/pubsyss2/xml/en/procedures/377/377-016-005.html) | 16                |
| Service                        | X15 CM2350<br>X116B | 5411186         | Flywheel        | Refer to Procedure 016-005 (/qs3/pubsyss2/xml/en/procedures/453/453-016-005.html) |                   |
| Service                        | X15 CM2350<br>X123B | 5504510         | Flywheel        | Refer to Procedure 016-005 (/qs3/pubsyss2/xml/en/procedures/538/538-016-005.html) |                   |
| Service                        | X15 CM2350<br>X129B | 5504557         | Flywheel        | Refer to Procedure 016-005 (/qs3/pubsyss2/xml/en/procedures/608/608-016-005.html) |                   |
| Service                        | X15 CM2450<br>X124B | 5504583         | Flywheel        | Refer to Procedure 016-005 (/qs3/pubsyss2/xml/en/procedures/564/564-016-005.html) |                   |

| Table 3, Publications Affected |                     |                 |                 |                                                                                     |                   |
|--------------------------------|---------------------|-----------------|-----------------|-------------------------------------------------------------------------------------|-------------------|
| Manual type                    | Engine              | Bulletin Number | Procedure Title | Procedure Number                                                                    | Procedure Section |
| Service                        | X15 CM2450<br>X134B | 5613373         | Flywheel        | Refer to Procedure 016-005 (/qs3/pubsyss2/xml/en/procedures/643/643-016-005.html)   |                   |
| Service                        | X15 CM2450<br>X142B | 6380843         | Flywheel        | Refer to Procedure 016-005 (/qs3/pubsyss2/xml/en/procedures/1014/1014-016-005.html) |                   |
| Service                        | X15 CM2670<br>X145B | 5676914         | Flywheel        | Refer to Procedure 016-005 (/qs3/pubsyss2/xml/en/procedures/1089/1089-016-005.html) |                   |
|                                |                     |                 |                 |                                                                                     |                   |

Document History

| Date       | Details        |
|------------|----------------|
| 2023-10-13 | Module Created |